Gavin Scoville

gavin.scoville@gmail.com
509.408.2425

EDUCATION

Western Washington University, Bellingham, WA
Bachelor of Arts - Economics August 2024

- **Minors — Mathematics, Environmental Science**
- GPA 3.94 — Honor Roll (4 years). Graduated with Honors: Cum Laude
  - Marine and Coastal Science Scholars Program
  - Economics Department Conference Travel Award
  - Economics Department Academic Excellence Award
  - Distinguished Graduate Award
- Course Highlights— *Stream Ecology (A), Topics in Marine and Coastal Science (A), Multivariate Statistics and Machine Learning (A), Advanced Environmental Economics (A), Mechanics of Materials (A), Multivariable Calculus and Analytical Geometry II (A)*

Research Abroad — Okavango Delta, Botswana, September 2023

- Worked in partnership with villages participating in Botswana's Community-Based Natural Resource Management (CBNRM) Program
- Collected biological field data in the Okavango Delta to quantify wildlife populations
- Studied ecology and conservation theory
- Presented a report on environmental patterns to local academics land managers

PUBLICATIONS, RECENT PROJECTS, AND PRESENTATIONS

- Performed analysis and played an integral part in developing the peer-reviewed paper:
  Glosser, D., Fish, N., **Scoville, G**., & Oos, K. (2026). Effect of temperature and atmosphere on the optical, thermal, and structural properties of fly ash particles. *Powder Technology*, *468*, 121650. https://doi.org/10.1016/j.powtec.2025.121650

- Identified patterns in global forest growth using machine learning techniques. Conducted in an internship with leading dendrochronologist Dr. Andy Bunn (see GitHub repository):

- Presented a lecture giving an overview of non-market valuation techniques to food-system researchers at the 2024 Agriculture Food and Human Values Society (AFHVS) international food systems conference
  Scoville, G., Plakias Z. (2024, June 7). *A Review of Quantitative Methods for Measuring Social Impacts of Food System Policies* [Conference presentation]. https://gavinscoville.github.io/GavinScoville/portfolio/AFHVS/

- Additional sample work available on my personal website:
  https://gavinscoville.github.io/GavinScoville/portfolio/

RESEARCH INTERESTS

I am passionate about innovating resilient adaptations to the Pacific Northwest's changing hydrology. Because the water cycle trickles through every aspect of our economy and ecology, this resource requires an interdisciplinary management approach. I am excited to apply my diverse background, interest in stochastic/chaotic systems, strong quantitative skills, and unending passion for creative problem solving to researching hydrological systems.

RESEARCH EXPERIENCE

Concentrated Solar Power Research Assistant
June 2025-Sep 2025

- Researched the usability of recycled coal fly ash for concentrated solar power systems
- Used the R programming language and machine learning methods to identify statistical patterns in laboratory data and produced publication quality data visualizations
- Contributed substantially to the development of the manuscript, which is now published in Powder Technologies

Estuarine Science Field and Lab Technician
May 2024-Sep 2024

- Operated scientific instruments to investigate carbon cycling and sedimentation rates in Pacific Northwest estuaries
- Conducted fieldwork, including navigating estuaries to collect precision elevation data and sediment cores
- Performed laboratory analyses and applied statistical techniques to quantify accretion rates and carbon content

Environmental Economics Research Assistant
April 2024-June 2024

- Reviewed literature for Dr. Plakias's Agricultural Economics Program at WWU
- Researched cutting-edge techniques for cost-benefit analysis of Environmental Projects
- Presented my findings at the AFHVS international conference in Syracuse, NY

Forest Ecology Data Science Internship
Jan 2024-Aug 2024

- Built machine learning algorithms for analyzing the International Tree Ring Database (IRTDB)
- Identified global patterns in how climate change is impacting forest growth rates

Okavango Delta Student Researcher
Sep 2023 - Dec 2023

- Worked in partnership with villages participating in Botswana's Community-Based Natural Resource Management (CBNRM) Program
- Collected biological field data in the Okavango Delta to quantify wildlife populations
- Evenings were spent studying ecology and conservation theory
- Presented a report on observed wildlife population patterns to local land managers.

Center for Economics and Business Research Research Assistant
Sep 2022 - Sep 2023

- Produced lifecycle analysis studies such as a report tracing recycling supply chains for Washington State
- Improved the design of our economic forecasts reports by automating data collection and building custom visualizations

MENTORSHIP AND COMMUNITY ENGAGEMENT

Math and Economics Tutoring
Sep 2022 - Sep 2023

- Identified and applied alternative teaching tools for teaching students with different learning styles
- Utilized socratic methods and problem based learning to motivate students
- Supported all of my students to pass their classes after working together

Community Engagement and Indigenous Partnerships Cohort
March 2023

- Participated in a six-week cohort on Indigenous–non-Indigenous partnerships focused on community engagement, truth, and reconciliation (White Swan Environmental)
- Gained training in how to form healthy Indigenous–non-Indigenous partnerships to address challenges such as resource management

Outdoor Leadership
Sep 2020 - Present

- Led dozens of trips introducing WWU students to camping, canoeing, skiing, climbing, and backpacking
- Empowered people to be comfortable navigating their public lands

RELEVANT SKILLS

Research and Data Skills: Handling large and messy time-series datasets for ecological, hydrological, and behavior modeling. Applying machine learning methods to identify complex patterns. Interpreting messy data to motivate decision making.

Technical/Programming: R programming language, Python, GitHub, STATA, SQL, GIS

Field/Lab Methods: Fluvial sedimentation analysis, discharge measurement, sediment coring, tension and compression testing, river flow monitoring, tree coring, riparian habitat assessments

Field Skills: Backcountry navigation, backpacking, climbing, equipment maintenance and repair, operating heavy equipment, operating motorized and non-motorized water vessels, first aid

ADDITIONAL PROFESSIONAL EXPERIENCE

Climbing Arborist Sep 2024 - Present

- Climbed trees & operated saws in challenging conditions with a diverse team of people
- Inspired clients to become effective stewards of their local landscape

Painter June 2022 - Sep 2022

- Exceeded productivity expectations while painting and remodeling houses

Line Cook June 2021 - Sep 2021

- Prepared dishes with a fast-paced team for an upscale Hawaiian restaurant
- Thrived under pressure, performing best when the restaurant depended on me

Farm Hand June 2020 - Sep 2020

- Raised livestock at Winzer Farms in Ellensburg, WA
- Operated heavy equipment such as an excavator, tractor, and a 1962 Ford dump truck